

Headline XVIII: the assembled stochastic posterior quotient

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Abstract

Unit 215 closes Astra #23's programme G: the paper's leading-order stochastic expectation expansion, conditional on the chart decomposition. For finitely many charts $a \in \kappa$ with their own dimensions and exponents, deterministic continuous observables φ_a and densities $c_a > 0$, a common β , jointly convergent random phases $X_n \Rightarrow Z$ in $\prod_a C([0, 1]^{d_a+1})$ and deterministic $N_n \rightarrow \infty$ ($N_n > 1$),

$$\frac{\sum_a \int \varphi_a c_a u^{h_a} e^{-\beta N_n^2 u^{2k_a} + \beta N_n u^{k_a} X_n^a}}{\sum_a \int c_a u^{h_a} e^{-\beta N_n^2 u^{2k_a} + \beta N_n u^{k_a} X_n^a}} \Rightarrow \frac{\sum_{a \in S} \text{phaseCoeff}_a(Z^a, \varphi_a c_a)}{\sum_{a \in S} \text{phaseCoeff}_a(Z^a, c_a)},$$

S the charts with minimal exponent $p_a = 2\lambda_a$ and, among those, maximal multiplicity (`headline_stochastic_assembled_posterior`). It is a ratio of sums, not a sum of chart ratios; the selected limiting denominator is a.s. (indeed surely) positive. Zero `sorry/axiom`; 9092 jobs.

1 The things formalised

```
theorem tendstoInDistribution_normChart_pairVec ... (g g : (a : ) → C(closedCube
  (d a + 1), )) :
  TendstoInDistribution (fun n a => (normChart ... (Nseq n) (X n a, g a),
    normChart ... (X n a, g a))) L
  (fun a => (limChart ... (Z a, g a), limChart ... (Z a, g a))) (fun _ =>
    ) '
theorem headline_stochastic_assembled_posterior [Fintype ] [Nonempty ] (d) (h k)
  (hk) (l) ( ) (hl) (h) (hmin) (hatt)
  (X : → Ω → PhaseSpaces d) (hXm) (Z) (hZm) (hX : TendstoInDistribution X L Z
    (fun _ => ) ')
  (Nseq) (hN : Tendsto Nseq L atTop) (hN1 : n, 1 < Nseq n) ( c) (hcpos : a
    x, 0 < c a x) :
  TendstoInDistribution (fun n => ( a, chartIntegral (d a + 1) (h a) (k a)
    (Nseq n) (extCube (X n a))
    (extCube ( a * c a))) / a, chartIntegral ... (extCube (X n a)) (extCube
    (c a))) L
  (fun _ => ( a univ.filter sel, limChart (h a) (k a) (l a) (Z a, a *
    c a))
    / a univ.filter sel, limChart (h a) (k a) (l a) (Z a, c a)) (fun
    _ => ) '

```

2 Mathematics

Per chart, $\sum_a I_a/S_* = \sum_a (S_a/S_*)(I_a/S_a)$ with $S_a = N^{-p_a}(\log N)^{m_a-1}$ and S_* the selected normalisation. The vector $(I_a^{\varphi c}/S_a, I_a^c/S_a)_a$ converges jointly in distribu-

tion (joint pair-vector Headline XVI: common phase vector, errors to 0 in probability collected by `tendstoInMeasure_pi_zero`); the deterministic ratios $S_a/S_* \rightarrow e_a \in \{0,1\}$ (`scale_ratio_tendsto`); Slutsky pairs the two; the continuous map $(v,w) \mapsto (\sum_a w_a v_a^{(1)}, \sum_a w_a v_a^{(2)})$ gives the assembled normalised numerator and denominator jointly; the selected denominator $\sum_{a \in S} \text{phaseCoeff}_a(Z^a, c_a)$ is a finite sum of strictly positive terms over a nonempty selection; the quotient theorem of unit 157 and the exact cancellation of S_* (all $N_n > 1$) finish. As Astra #23 insisted, this is the honest random object — the denominator weights are random and do not cancel chart by chart.

3 Lean notes

- The whole proof compiled at the first attempt; the only friction was cosmetic (five long lines, `continuous_finset_sum` deprecated for `continuous_finsetSum`).
- `scale_ratio_tendsto` for the abstract $(p, m) = (2\lambda_a, \text{multCount}_a)$ is accepted against the `leadScale` form by `defeq` after `tendsto_pi_nhds`; no rewriting needed.
- Passing the summation map `g` through `continuous_comp` and then reading off the two components with `Measurable.fst/snd` avoids restating measurability of the sums.
- `boole_mul` with `Finset.sum_filter` converts $\sum_a e_a f_a$ into the filtered sum.

4 Status

Programme G complete (units 214–215). Remaining per Astra #23: D2 (signed orthants with phase, if the paper’s interior charts stay unsplit) and the hand-off report v3.